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Semiconductor device

Abstract

A semiconductor device includes case resin having frame part defining space in which semiconductor chip is disposed and bottom part located under the frame portion; external connection terminal having external terminal partially embedded in the frame part, and internal terminal disposed on the bottom part to extend from the external terminal into the space; wire electrically connecting the semiconductor chip and the internal terminal; and sealing resin formed in the space to cover the semiconductor chip, the wire, and the internal terminal. The internal terminal has rectangular connecting portion, and step portions disposed at opposite ends of the connecting portion, respective portions of the upper surfaces of the step portions facing each other are covered with vibration controlling portions implemented by the case resin, and the sealing resin is filled in first grooves exposing other portions of the upper surfaces of the step portions.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. P2022-021954 filed on Feb. 16, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a semiconductor device and more particularly a power semiconductor device.

2. Description of the Related Art

(3) In a power semiconductor device, such as a power converter, a switching device and the like, a

circuit board on which semiconductor elements are arranged is housed in a resin case integrally formed with external connection terminals by insert molding. The case is integrally formed with the external connection terminals including power terminals for power input/output and signal terminals for signal input/output by insert molding. When the semiconductor device operates, the semiconductor elements generate heat. As the operation time of the semiconductor device becomes longer, deterioration, such as cracks in the bonding materials for bonding the semiconductor elements and the power terminals to the circuit board and deformation of the wires bonded to the semiconductor elements occurs. To prevent such deterioration, the semiconductor elements inside the case are sealed with resin.

(4) Since the difference in the linear expansion coefficient between the sealing resin and the case material is large, the sealing resin may be peeled off from the case material due to occurrence of repeated stress in the heat cycle by heat generation of the semiconductor elements and temperature change in the external atmosphere. Therefore, reliability may be reduced due to disconnection of wires bonded to the semiconductor elements or moisture absorption from the gap between the case material and the sealing resin.

(5) JP 2017-199818 A describes improving the adhesion between the case material and the sealing resin with grooves disposed on both sides of the external connecting portion of the external connection terminal along the periphery of the bottom of the case, and preventing degradation in the bonding property with the vibration controlling portion disposed in the groove adjacent to the internal connecting portion. JP 2000-349219 A discloses that the step portion is formed at the end of the external connection terminal apart from the wire bonding portion to secure the area of the wire bonding portion and improve adhesion between the external connection terminal and the case resin. JP 2016-111028 A discloses that the case material is placed in the recessed portion formed at the tip of the internal connecting portion to fix the internal connecting portion, to prevent the occurrence of a gap in the lower portion of the internal connecting portion at the time of tie-bar cutting of the insert-molded lead frame. JP 2019-67885 A discloses that the recessed portion or the bent portion is formed on the back side of the internal connecting portion to improve the degree of adhesion between the case material and the internal connecting portion and to secure the area of the wire bonding portion. WO 2015/152373 describes exposing the wire bonding portion in the recess of the case material disposed at the tip of the internal connecting portion to prevent the sealing resin and the external connection terminal from peeling off and to prevent wire disconnection.

(6) Conventionally, in order to improve the adhesion between the case resin and the sealing resin, a contact surface of the case resin are addressed by application of coating material or roughening. However, the application of the coating material increases the manufacturing cost due to addition of the material and manufacturing processes. The coating material may be difficult to apply depending on the internal shape of the case. To roughen the contact surface, it is necessary to roughen the inner surface of the molding die by electric discharge machining or the like. Wear of the inner surface of the mold lowers surface roughness and deteriorates adhesion. Therefore, the management and maintenance of the molding die are required, and the manufacturing cost increases.

SUMMARY OF THE INVENTION

(7) An aspect of the present invention inheres in a semiconductor device, including: (a) a case resin having a frame part defining a space in which a semiconductor chip is disposed, and a bottom part located under the frame portion; (b) an external connection terminal having an external terminal partially embedded in the frame part, and an internal terminal disposed on the bottom part so as to extend from the external terminal into the space; (c) a wire electrically connecting the semiconductor chip and the internal terminal; and (d) a sealing resin formed in the space to cover the semiconductor chip, the wire, and the internal terminal, wherein, the internal terminal has a rectangular connecting portion in plan view, and step portions disposed at opposite ends of the connecting portion in parallel with an extending direction of the internal terminal into the space,

upper surfaces of the step portions are lower than an upper surface of the connecting portion, respective portions of the upper surfaces of the step portions facing each other are covered with vibration controlling portions implemented by the case resin, and the sealing resin is filled in first grooves exposing other portions of the upper surfaces of the step portions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic plan view illustrating an example of a semiconductor device according to an embodiment of the present invention;

(2) FIG. **2** is a schematic cross-sectional view taken along line A-A in FIG. **1**;

(3) FIG. **3** is a circuit diagram illustrating an example of a semiconductor device according to an embodiment of the present invention;

(4) FIG. **4** is an enlarged view of portion B in FIG. **1**;

(5) FIG. **5** is a schematic cross-sectional view taken along the line C-C in FIG. **4**;

(6) FIG. **6** is a schematic cross-sectional view taken along the D-D direction in FIG. **4**;

(7) FIG. **7** is a schematic plan view illustrating another example of a semiconductor device according to an embodiment of the present invention;

(8) FIG. **8** is a schematic cross-sectional view taken along the line E-E in FIG. **7**; and

(9) FIG. **9** is a schematic cross-sectional view taken along the line F-F in FIG. **7**.

DETAILED DESCRIPTION

(10) Hereinafter, embodiments of the present invention will now be described with reference to the drawings. In the description of the drawings, the same or similar parts are denoted by the same or similar numerals, and duplicate descriptions are omitted. However, the drawings are schematic, and the relationship between the thickness and the plane dimension, the ratio of the thickness of each layer, and the like may be different from the actual ones. Also, portions having different dimensional relationships and ratios may be included among the drawings. Further, the following embodiments exemplify apparatuses and methods for embodying the technical idea of the present invention, and the technical idea of the present invention does not specify the material, shape, structure, arrangement, etc. of the components as follows.

(11) Further, the definition of the vertical direction or the like in the following description is merely a definition for convenience of explanation and does not limit the technical idea of the present invention. For example, in the following description, “top” and “bottom” in the “top conductive layer” and “bottom conductive layer” are only choices for convenience and are not defined with respect to the direction of gravity of the earth. Therefore, it is needless to say that when the object is rotated by 90° and observed, the top and bottom are converted to the right and left and read, and when the object is rotated by 180° and observed, the top and bottom are inverted and read.

(12) In the following description, a three-phase inverter circuit will be used as a representative example of a semiconductor device, but the semiconductor device of the present invention is not limited to the three-phase inverter circuit, and may be a power module such as a full bridge circuit or a half bridge circuit. Although an insulating gate bipolar transistor (IGBT) is used as the switching transistor of the inverter circuit, the description is not limited. For example, it may be a bipolar device such as an electrostatic induction thyristor (SI thyristor) or a gate turn-off (GTO) thyristor, or it may be a MOS field effect transistor (FET), a MISFET, an electrostatic induction transistor (SIT), or the like. As the reflux diode D of the inverter circuit, a fast barrier diode (FRD), a Schottky barrier diode (SBD), or the like can be employed.

Embodiments

(13) As illustrated in FIGS. **1** and **2**, a semiconductor device according to an embodiment of the present invention is a semiconductor power module encompassing a three-phase inverter circuit

housed in a case resin **1**. The case resin **1** has a frame part **1***a* defining respective spaces in which leg circuits **30***u*, **30***v*, **30***w* implementing a 3-phase inverter circuit are disposed, and a bottom part **1***b* located in a bottom of the frame part **1***a*. The bottom part **1***b* extends from the frame part **1***a* to the space defined by the frame part **1***a*. The case resin **1** has terminals including external connection terminals (**3**E, **4**E), (**3**G, **4**G), (**3**F, **4**F), (**3**H, **4**H), positive terminals **15**, negative terminals **16**, and output terminals **17**. The terminals are integrally formed by insert molding, with a metallic lead frame into the case resin **1**. The external connection terminal (**3**E, **4**E) has an external terminal **3**E embedded and fixed in the frame part **1***a*, and an internal terminal **4**E extending from the external terminal **3**E into the space defined by the frame part **1***a* and disposed on the bottom part **1***b*. The external connection terminal (**3**G, **4**G) has an external terminal **3**G embedded and fixed in the frame part **1***a*, and an internal terminal **4**G extending from the external terminal **3**G into the space defined by the frame part **1***a* and disposed on the bottom part **1***b*. The external connection terminal (**3**F, **4**F) has an external terminal **3**F embedded and fixed in the frame part **1***a*, and an internal terminal **4**F extending from the external terminal **3**F and disposed on the bottom part **1***b*. The external connection terminal (**3**H, **4**H) has an external terminal **3**H embedded and fixed in the frame part **1***a*, and an internal terminal **4**H extending from the external terminal **3**H and disposed on the bottom part **1***b*.

(14) The case resin **1** is fixed to a cooling plate **21** by an adhesive **20**. The case resin **1** is formed, for example, of an insulating thermoplastic resin. For the thermoplastic resin, polyphenylene sulfide (PPS) resin, polybutylene terephthalate (PBT) resin, polybutylene succinate (PBS) resin, polyamide (PA) resin, acrylonitrile butadiene styrene (ABS) resin and the like may be suitably used.

(15) The respective leg circuits **30***u*, **30***v*, **30***w* include semiconductor chips **2***a*, **2***b* mounted on respective insulated circuit boards **10**. As illustrated in FIG. **2**, each of the insulated circuit boards **10** includes an insulating plate **11**, conductor layers **12***a*, **12***b*, **12***c* patterned on an upper surface of the insulating plate **11**, and a conductor layer **13** disposed on a lower surface of the insulating plate **11**. As illustrated in FIG. **1**, the semiconductor chip **2***a* is disposed on the conductor layer **12***a*, and the semiconductor chip **2***b* is disposed on the conductor layer **12***b*. The insulated circuit board **10** is joined to the cooling plate **21** via the conductor layer **13** by a jointing material such as solder. The insulated circuit board **10** may be, for example, a direct copper bonding (DCB) substrate in which copper is eutectically bonded to the surface of a ceramic substrate, an AMB substrate in which a metal is arranged on the surface of the ceramic substrate by an active metal brazing (AMB) method, or the like. The material of the ceramic substrate may be, for example, silicon nitride (Si_3N_4), aluminum nitride (AlN), alumina (Al_2O_3) or the like.

(16) As illustrated in FIG. **2**, sealing resins **9**, which are omitted in FIG. **1**, are formed in the spaces defined by the frame parts **1***a* of the case resin **1** so as to cover the insulated circuit boards **10** on which the semiconductor chips **2***a*, **2***b* are mounted, wires **7**, and the internal terminals **4**G, **4**E, **4**F, **4**H. Although epoxy resin is used for the sealing resins **9**, other resin may be used. For example, a thermoplastic resin with excellent insulating and adhesive properties, such as silicone resin, urethane resin, polyimide resin, polyamide resin, polyamideimide resin, and the like, which are different from case resin **1**, can be adopted.

(17) In each of the leg circuits **30***u*, **30***v*, **30***w*, for example, an upper arm is implemented with the semiconductor chip **2***a* and a lower arm is implemented with the semiconductor chip **2***b*, as illustrated in FIG. **3**. Each of the semiconductor chips **2***a*, **2***b* has a switching transistor T made of an IGBT and a freewheeling diode D connected in antiparallel to the switching transistor T. A cathode electrode K of the freewheeling diode D is electrically connected to a collector electrode C of the switching transistor T. An anode electrode A of the freewheeling diode D is electrically connected to an emitter electrode E of the switching transistor T. The collector electrodes C in the semiconductor chips **2***a* are electrically connected to the positive terminals **15**, respectively. The emitter electrodes E in the semiconductor chips **2***b* are electrically connected to the negative

terminals **16**, respectively. Each emitter electrode E in the semiconductor chips **2***a* and each collector electrode C in the semiconductor chips **2***b* are electrically connected to the output terminals **17**, respectively.

(18) As illustrated in FIG. **1**, a gate electrode G of each switching transistor T in the semiconductor chips **2***a* is electrically connected to the internal terminal **4**G via the wire **7**. An auxiliary emitter electrode (not shown) electrically connected to each emitter electrode E in the semiconductor chips **2***a* is electrically connected to the internal terminal **4**E via the wire **7**. A gate electrode G of each switching transistor T in the semiconductor chips **2***b* is electrically connected to the internal terminal **4**H via the wire **7**. An auxiliary emitter electrode (not shown) electrically connected to each emitter electrode E in the semiconductor chips **2***b* is electrically connected to the internal terminal **4**F via a wire **7**. A resistance element for adjusting a switching speed and a loss may be added to the gate electrode G. The auxiliary emitter electrode is an auxiliary electrode for detecting a voltage or the like on the emitter electrode side.

(19) In each of the leg circuits **30***u*, **30***v*, and **30***w*, as illustrated in FIG. **3**, a connection node between the emitter electrode E of the semiconductor chip **2***a* and the collector electrode C of the semiconductor chip **2***b* are electrically connected to the output terminal **17**. For example, during operation of the semiconductor device according to the embodiment, a positive electrode P and a negative electrode N of an external DC power source (not shown) are connected to the positive terminals **15** and the negative terminals **16** of the leg circuits **30***u*, **30***v*, **30***w*, respectively. In each of the leg circuits **30***u*, **30***v*, **30***w*, a pulse signal is fed from a control circuit (not shown) to the respective gate electrodes G via the external terminals **3**G, **3**H, and a reference signal is fed to the respective emitter electrodes E via the external terminals **3**E, **3**F. As a result, for example, U-phase AC power is supplied from the output terminal **17** of the leg circuit **30***u*, V-phase AC power is supplied from the output terminal **17** of the leg circuit **30***v*, and W-phase AC power is supplied from the output terminal **17** of the leg circuit **30***w*. And thus, 3-phase AC power is supplied to a load **50** such as a motor.

(20) FIG. **4** is an enlarged view of portion B illustrated in FIG. **1**, and illustrates configurations of the external connection terminals (**3**E, **4**E), (**3**G, **4**G) integrally molded with the case resin **1**. In FIG. **4**, the sealing resin **9** is omitted from illustration. FIG. **5** is a schematic cross-sectional view perpendicularly taken from line C-C in FIG. **4** passing through the bonding end of the wire **7**, and FIG. **6** is a schematic cross-sectional view perpendicularly taken from line D-D in FIG. **4** avoiding the bonding end of the wire **7**. Note that the external connection terminals (**3**F, **4**F), (**3**H, **4**H) have the same configuration as the external connection terminals (**3**E, **4**E), (**3**G, **4**G). As illustrated in FIGS. **4** to **6**, each of the internal terminals **4**E, **4**G of the external connection terminals (**3**E, **4**E), (**3**G, **4**G) includes a connecting portion **4***a* and step portions **4***b*. The connecting portion **4***a* has a rectangular shape in a plan view. The step portions **4***b* are disposed at opposite ends of the connecting portion **4***a* in parallel with an extending direction of the internal terminal **4**E or **4**G from the external terminal **3**E or **3**G into the space where the semiconductor chips **2***a*, **2***b* are disposed. Each of the step portions **4***b* has a width Ws, and each upper surface of the step portions **4***b* is lower than an upper surface of the connecting portion **4***a* by a depth Ds. The upper surface of the connecting portion **4***a* is planar at a level substantially the same height as the upper surface of the bottom part **1***b*.

(21) As illustrated in FIG. **4**, grooves (first grooves) **6** exposing the upper surfaces of the step portions **4***b* are located between the bottom part **1***b* and the connecting portion **4***a* in adjacent sides to the frame part **1***a* and in tip sides of the internal terminals **4**E, **4**G. The grooves **6** are separately disposed, and have lengths Lc in the adjacent sides to the frame part **1***a* and lengths Lt in the tip sides of the internal terminals **4**E and **4**G, for example. As illustrated in FIG. **6**, the sealing resin **9** is filled in the grooves **6** where the upper surfaces of the step portions **4***b* are exposed. For example, the PPS resin used for the case resin **1** and the epoxy resin used for the sealing resin **9** have different linear expansion coefficients and have poor adhesion. In a conventional semiconductor

device, the sealing resin made of epoxy resin may be easily peeled off from the case resin made of PPS resin due to occurrence of repeated stress caused by heat generation of the semiconductor chip. Therefore, breaking of the wire connecting the semiconductor chip to the external connection terminal and moisture absorption from the gap between the case resin and the sealing resin are likely to occur. In the semiconductor device according to the embodiment, the grooves **6** are disposed in the bottom part **1***b* of the case resin **1**, and the sealing resin **9** is filled in the grooves **6** in which the upper surfaces of the step portions **4***b* are exposed. Therefore, the sealing resin **9** is into contact with the metal surfaces of the step portions **4***b*. The step portion **4***b* made of metal is superior to the case resin **1** in adhesion with the sealing resin **9**. Therefore, the adhesion between the sealing resin **9** and the case resin **1** can be improved. Additionally, since the sealing resin **9** is filled in the groove **6**, the adhesion can be further improved by the anchor effect.

(22) Further, as illustrated. FIG. **4**, vibration controlling portions **5** are defined between the grooves **6** to selectively cover the upper surfaces of the step portions **4***b* with the case resin **1**. The vibration controlling portions **5** are provided with a length Lv. As illustrated in FIG. **5**, both ends of the connecting portions **4***a* of the internal terminals **4**E, **4**G are fixed so as to fix the step portions **4***b* by the vibration controlling portions **5** defined in the bottom part **1***b* of the case resin **1**. Since the wires **7** are bonded to the upper surfaces of the connecting portions **4***a* by ultrasonic bonding, there is a concern about degradation in the bondability of the wire due to the vibration of the ultrasonic waves applied during bonding. In the semiconductor device according to the embodiment, the vibration of the connecting portions **4***a* during ultrasonic bonding is controlled by fixing the connecting portions **4***a* via the vibration controlling portions **5** disposed on the step portions **4***b* of the internal terminals **4**E, **4**G. As illustrated in FIG. **4**, in order to control the vibration more effectively, it is preferable to join the wire **7** so that the bonding end of the wire **7** is placed in a bonding area **8** defined by the opposing vibration controlling portions **5**. In a plan view, the bonding area **8** has the same length Lv as the vibration controlling portion **5** in the extending direction from the external terminals **3**E, **3**G to the space defined by the frame part **1***a*. The length Lv of the bonding area **8** is longer than the length Lb of the bonding end of the wire **7**. Thus, by including the bonding end of the wire **7** in the bonding area **8**, the longitudinal and lateral vibrations of ultrasonic waves applied at the time of wire bonding of the wire **7** can be controlled.

(23) In the above description, in order to improve the adhesion between the sealing resin **9** and the case resin **1**, the grooves **6** are disposed in the bottom part **1***b* of the case resin **1**, but the grooves may be further disposed in the frame part **1***a*. For example, FIG. **7** is an enlarged view corresponding to the portion B illustrated in FIG. **1**, FIG. **8** is a schematic cross-sectional view perpendicularly taken from line E-E in FIG. **7**, and FIG. **9** is a schematic cross-sectional view perpendicularly taken from line F-F in FIG. **7**. As illustrated in FIGS. **7** to **9**, grooves (second grooves) **6***a* are disposed on the side wall of the frame part **1***a* of the case resin **1**. The metal surfaces of the external terminals **3**E, **3**G are partially exposed in the grooves **6***a*, and the sealing resin **9** filled in the grooves **6***a* is into contact with the metal surfaces of the external terminals **3**E, **3**G. Therefore, the adhesion between the sealing resin **9** and the case resin **1** can be improved. Additionally, since the sealing resin **9** is filled in the grooves **6***a*, the adhesion can be further improved by the anchor effect. In addition, insert molding is executed multiple times, for example, twice, to form the grooves **6***a* on the side wall of the frame part **1***a*. Only the part corresponding to the frame part **1***a* in which lead frames for the external connection terminals are embedded is molded by the primary insert molding to form the grooves **6***a* on the side wall of the part corresponding to the frame part **1***a*. Next, the remaining part including the bottom part **1***b* is molded by secondary insert molding to form the case resin **1**.

OTHER EMBODIMENTS

(24) Although the present invention has been described in accordance with the above-disclosed embodiments, it should not be understood that the statements and drawings forming part of this disclosure are intended to limit the present invention. It should be appreciated that the disclosure of

the description and drawings of the present invention will reveal various alternative embodiments, embodiments, and operational techniques to those skilled in the art. Moreover, the present invention naturally includes various embodiments not described herein, such as a configuration in which the respective configurations described in the above embodiments and the respective modifications are optionally applied. Therefore, the technical scope of the invention is determined only by the matters specifying the invention in the claims, which are reasonable from the above exemplary description.

Claims

1. A semiconductor device comprising: a case resin having a frame part defining a space in which a semiconductor chip is disposed, and a bottom part located under the frame portion; an external connection terminal having an external terminal partially embedded in the frame part, and an internal terminal disposed on the bottom part so as to extend from the external terminal into the space; a wire electrically connecting the semiconductor chip and the internal terminal; and a sealing resin formed in the space to cover the semiconductor chip, the wire, and the internal terminal, wherein, the internal terminal has a rectangular connecting portion in plan view, and step portions disposed at opposite ends of the connecting portion in parallel with an extending direction of the internal terminal into the space, upper surfaces of the step portions are lower than an upper surface of the connecting portion, respective portions of the upper surfaces of the step portions facing each other are covered with vibration controlling portions implemented by the case resin, and the sealing resin is filled in first grooves exposing other portions of the upper surfaces of the step portions.
2. The semiconductor device according to claim 1, wherein a bonding end of the wire on the connecting portion is placed in a bonding area defined by the opposing vibration controlling portions.
3. The semiconductor device according to claim 1, wherein the frame part is provided with a second groove in which a surface of the external terminal is partially exposed and the sealing resin is filled.
4. The semiconductor device according to claim 1, wherein the first grooves are separately disposed across the vibration suppression portion in the extending direction.